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Wind and Solar Radiation Complementarity Analysis

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Abstract

to be completed

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Introduction

This short report aims at reproduce [Cauz et al. \[2025\]](#)'s results on the complementarity between electricity production from solar panels and wind turbines. Overall, they find a negative correlation between the two energy sources, mitigating risks to which renewables energy sources are exposed to.

The conclusions of this work require cautious interpretation since, since I do not use electricity production data as [Cauz et al. \[2025\]](#). Rather I collect wind and solar radiation data at the origin of both wind and solar electricity.

1 Data and sample construction

Authors rely on real production data from 2020 to 2023 for wind and PV production in Switzerland from SFOE and ENTSO-E. I use data from [MetoSwiss](#) relying on measures of wind speeds and solar radiations across the country. These extensive data (10-minute slice) allow us to perform a complementarity analysis on the model of [Cauz et al. \[2025\]](#) work at hourly, daily, monthly and yearly level.

In the [code](#) pipeline, I first collect the data on MeteoSwiss website at hourly, daily, monthly and yearly level from 2020 to 2029 (because data include forecast). I focus solely on mean wind speed and solar radiation, both represented by a single variable². This variable choice constitute a first hypothesis of this work, I defend it in another [document](#). In short, electricity production from wind turbine is due to wind speed and wind speed depends on the area, obstacles, mountain corridor, temperature, *etc.* While solar radiation is the principle source of electricity production for PV. PV production depends also on its orientation (if we suppose a non-moving PV installation).

2 Measuring resource complementarity

As I aiming at reproduce [Cauz et al. \[2025\]](#)'s results, I focus on 2020–2023 period. I could extend it for eventual robustness tests [to move to robustness check section](#). Using hourly, monthly and yearly data, I test the [Spearman correlation](#) to produce comparable results. For addition checks, I could run a Pearson (standard) correlation [to move to robustness check section](#). My [results](#) are the following + [add the comparision of Cauz and al. \(2025\) paper](#)

- Hourly correlation: 0.183
- Daily correlation: [to be reported](#)
- Monthly correlation: 0.037
- Yearly correlation: -0.2

²We can note that I use directly available hourly, monthly and yearly data. I do not compute them myself using 10-minute slices measures.

In a second part of this work I specify my findings using minimum and maximum thresholds. Indeed, a wind turbine start to produce electricity with a wind speed, on average, between 3 m/s and 5 m/s and stops (in case of emergency) at 25 m/s. While a solar panel requires at least a radiation of 100 w/m^2 .

3 Preliminary evidence

A first remark to make is that we computed correlations on 128 stations out of 158 because of missing values. A part of omitted stations may have a combination of radiation and wind speed which could have favour results nearer to [Cauz et al. \[2025\]](#). Also we are studying correlations for all wind and radiation levels during every hours, night and day. We could think that during night hours the studied correlation would have been lower for all our measures.

4 Robustness checks

5 Conclusion

References

Marine Cauz, Nicolas Wyrsh, Lionel Perret, Christophe Ballif, and Alejandro Pena-Bello. Embracing wind power in the solar pv-dominated swiss landscape. *Energy Reports*, 2025. doi: 10.1016/j.egyr.2025.02.059.